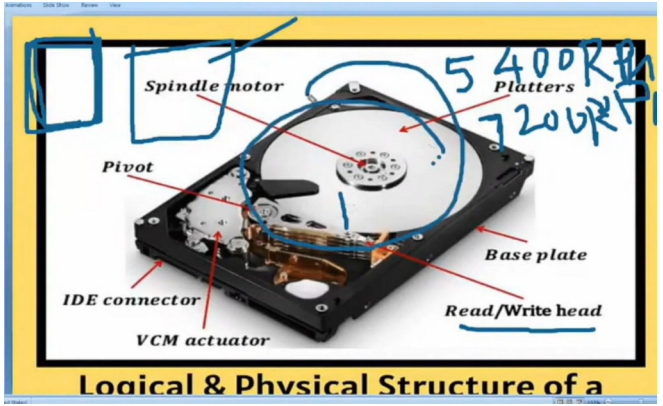
HARD DISK

Structure, working, types, partitions, file systems and disk management – explained step by step



hard disk

FROM THE LECTURE

- what is hard disk
- type of hard disk
- disk management
- (basic disk vs dynamic disk)
- partitions
- (primary, secondary /extended and logical)
- partition table

FROM THE LECTURE

- (mbr vs gpt)
- formate
- file system
- (fat, ntfs and refs)
- tracks & sectors
- resize disk
- (shrink and extend)

Topic guide – every topic in short

1 What is hard disk

Permanent magnetic storage of the PC.

2 Type of hard disk

HDD, SSD, SSHD and by interface and size.

3 Disk management

Windows tool to manage disks: Basic or Dynamic.

4 Partitions

Splitting a disk: primary, extended and logical.

5 Partition table

MBR or GPT: the map of partitions.

6 Format and file system

Prepare a partition: FAT, NTFS, ReFS.

7 Tracks & sectors

How data is laid out on the platter.

8 Resize disk

Shrink or extend a volume.

9 Working and uses

How it reads and writes, and where it is used.

Every topic in detail (1)

1 what is hard disk

The permanent storage of the computer. It keeps the OS, programs and files even when the power is off.

2 type of hard disk

HDD (spinning disks), SSD (flash chips) and hybrid SSHD, connected by IDE, SATA, SAS or NVMe.

3 disk management

The Windows tool to create, delete, format and resize partitions and to set a disk as basic or dynamic.

4 (basic disk vs dynamic disk)

Basic = normal partitions, used by most PCs. Dynamic = flexible volumes that can span or mirror disks.

Every topic in detail (2)

1 partitions

A disk is divided into parts so that C:, D: and E: can be separate, like rooms in a house.

2 (primary, secondary / extended and logical)

Primary can boot an OS. Secondary (extended) is a container. Logical drives live inside it.

3 partition table

A map at the start of the disk that records where each partition begins and ends.

4 (mbr vs gpt)

MBR = old, up to 2 TB, 4 primary partitions. GPT = new, over 2 TB, up to 128 partitions, UEFI.

Every topic in detail (3)

1 formate

Prepares a partition with a file system so files can be stored. It erases the existing data.

2 file system

The method the OS uses to name, store and find files on the disk.

3 (fat, ntfs and refs)

FAT = simple, for pen drives. NTFS = standard Windows. ReFS = resilient, for servers.

4 tracks & sectors

The disk surface is split into circular tracks, and each track into sectors – the smallest unit of storage.

Every topic in detail (4)

1 **resize disk**

Change the size of a partition without reinstalling Windows.

2 **(shrink and extend)**

Shrink frees space from a volume. Extend adds free space to a volume.

What is a hard disk?

1 Storage device

A hard disk drive (HDD) is the permanent storage of the computer – the secondary memory.

2 Non-volatile

Data stays saved even when the power is off, unlike RAM.

3 Stores everything

Operating system, programs, documents, photos, music and videos.

4 Magnetic storage

Data is written as tiny magnetic patterns on spinning metal disks called platters.

Hard disk at a glance

1 Capacity

Measured in GB and TB. Home disks are commonly 500 GB to several TB.

2 Speed (RPM)

5400 or 7200 revolutions per minute, as noted in the lecture.

3 Interface

Connects with IDE (old) or SATA (modern) cable.

4 Cache

A small fast buffer (a few MB to 256 MB) that speeds up transfers.

5 Form factor

3.5-inch for desktops, 2.5-inch for laptops and portable drives.

6 Handle with care

Moving parts: avoid shocks and drops, especially while running.

Advantages and limits of a hard disk

1 Advantages

Very large capacity, low cost per GB, long-term storage, easy to replace.

2 Limits

Slower than SSD, moving parts can fail, uses more power, makes noise.

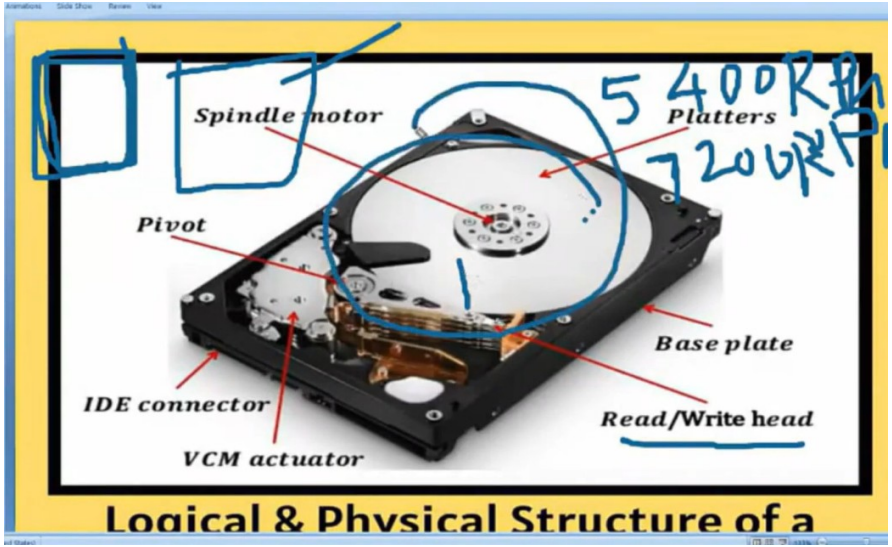
3 Good for

Bulk storage: movies, backups, archives, servers.

4 Not ideal for

Speed-critical work such as running the OS (an SSD is better).

What is HDD?



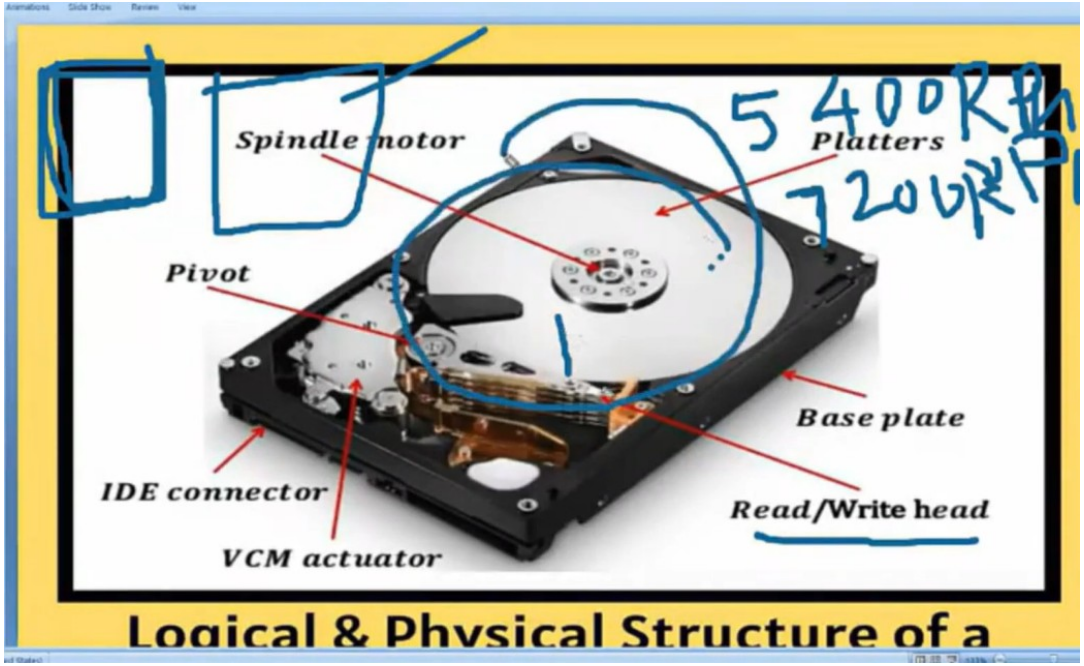
HDD = Hard Disk Drive
The traditional storage device of a computer.

Mechanical
Uses spinning metal platters and a moving read/write head.

Magnetic storage
Data is saved as magnetic patterns on the platters.

Speed in RPM
Platters spin at 5400 or 7200 RPM. Higher RPM = faster access.

Logical & physical structure



Spindle motor: Spins the platters

Platters: Magnetic disks that hold data

Pivot: Axis the arm swings on

VCM actuator: Moves the arm

Read/Write head: Reads and writes data

Base plate: Metal body of the drive

IDE connector: Connects to the board

Hard disk parts – Explained

1 Platters

Round, polished disks coated with magnetic material. Data lives on their surfaces.

2 Spindle motor

Spins the platters at a constant speed: 5400 RPM or 7200 RPM (noted in the lecture).

3 Read/Write head

Floats just above the platter and reads or writes the magnetic data.

4 VCM actuator and pivot

The voice coil motor swings the arm on the pivot to reach the right track.

5 Base plate

The sealed metal base that holds all parts safely.

6 IDE connector

Older interface that links the disk to the motherboard for data and power.

How a hard disk works

1. Power on

The spindle motor spins the platters up to 5400 or 7200 RPM.

2. Request

The CPU asks for a file; the controller finds its track and sector.

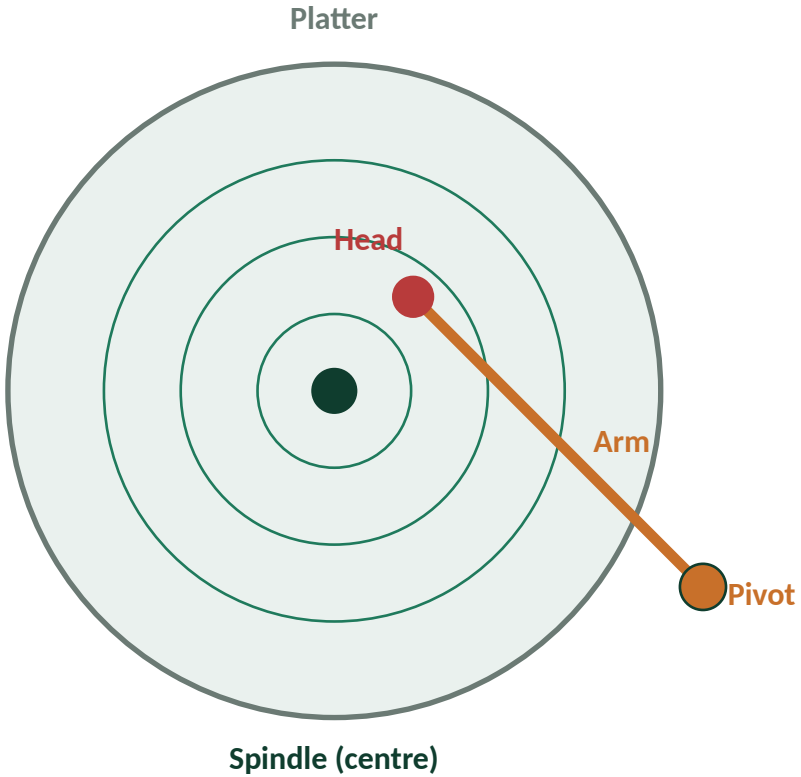
3. Move

The VCM actuator swings the head over the correct track.

4. Read / write

The head senses or changes the magnetic pattern, and data goes to RAM.

Working diagram



Top view of a drive

The platter spins around the spindle. The arm swings on the pivot so the head reaches any circular track. The head never touches the platter; it floats on a thin cushion of air.

Spin + swing

Spinning brings the right sector under the head, and swinging chooses the right track.

Reading and writing data

1 Seek time

Time the arm takes to move the head to the right track.

2 Rotational latency

Wait for the right sector to spin under the head; a 7200 RPM disk waits less than 5400 RPM.

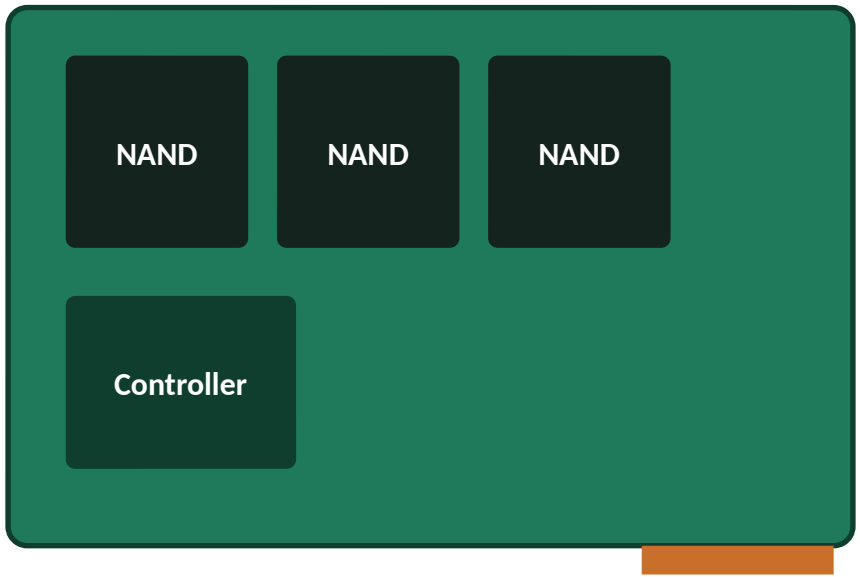
3 Transfer rate

How fast data flows once the head is in place (MB/s).

4 Why RPM matters

Higher RPM = faster access, but more heat, noise and power use.

What is SSD?



Flash chips + controller on a circuit board

SSD = Solid State Drive
Storage with no moving parts.

Flash memory
Data is kept in NAND flash chips as electrical charges.

Controller
A small chip manages reading, writing and wear.

Fast and quiet
Loads Windows and programs much faster than an HDD.

HDD vs SSD

	HDD	SSD
Full form	Hard Disk Drive	Solid State Drive
Technology	Magnetic spinning platters	Flash (NAND) memory chips
Moving parts	Yes: motor, platters, head	None
Speed	Slower (about 80–160 MB/s)	Much faster (500 MB/s and above)
Noise	Spinning and clicking sounds	Silent

HDD vs SSD – continued

	HDD	SSD
Shock resistance	Weak: a drop can damage the head	Strong: no moving parts
Price per GB	Cheaper	More expensive
Capacity	Very large (several TB)	Growing, but costly at high sizes
Power and heat	Uses more power	Uses less power
Best for	Bulk storage and backup	OS, programs, laptops, speed

Which one to choose?

1 Choose HDD when

You need a lot of space at low cost: movies, backups, CCTV and archives.

2 Choose SSD when

You want speed: faster boot, quick apps and a responsive laptop.

3 Best of both

Install the OS on an SSD and keep big files on an HDD. SSHD hybrids try to combine both.

4 Remember

Both are non-volatile secondary memory. They differ in how they store data.

Types of hard disk

1 **HDD**

Mechanical drive with spinning platters. Large and cheap, but slower.

2 **SSD**

Solid state drive with flash chips and no moving parts. Fast and silent.

3 **SSHD**

Hybrid: an HDD with a small SSD cache for speed.

4 **External drive**

An HDD or SSD in a case, connected by USB. Used for backup and carrying data.

Types by interface

1 IDE / PATA

Old parallel interface with a wide ribbon cable and a 40-pin port.

2 SATA

Modern serial interface with a thin cable; used by most HDDs and SSDs.

3 SCSI

Older fast interface used on servers.

4 SAS

Newer serial version of SCSI for servers.

5 NVMe

Very fast SSDs that connect through PCI Express, such as M.2 drives.

6 USB

External drives connect through the USB port.

Uses of hard disk

1 Operating system

Windows and all installed software are stored and loaded from the disk.

2 Personal data

Documents, photos, music and videos are kept on it.

3 Backup

External hard disks keep copies of important data.

4 Servers

Databases, websites and file sharing store data on large disks.

5 CCTV / DVR

Recordings from cameras are saved on surveillance hard disks.

6 Media and gaming

Large games, video editing files and film libraries need big storage.

Hard Disk Companies

Samsung

South Korean company, known for SSDs and storage.

Segate

Written as in the lecture. Seagate is a major HDD maker.

Toshiba

Japanese company making HDDs and flash storage.

WDC

Western Digital Corporation, a leading HDD brand.

Transcend

Taiwan-based brand of SSDs, pen drives and memory cards.

Disk Management

Basic

The default disk type. Uses normal partitions (primary, extended, logical). Used on most PCs.

Dynamic

Uses flexible volumes that can span or mirror disks. Windows only.

Disk Management – Explained

	Basic disk	Dynamic disk
Partitions	Primary, extended and logical partitions	Volumes (simple, spanned, striped, mirrored)
Use	Default; used on most PCs	Advanced: join disks or add fault tolerance
Resize	Limited	Easy to extend across disks
Support	All systems	Windows only

Opening Disk Management

1 Open

Press Windows + R, type diskmgmt.msc, press Enter. Or right-click This PC → Manage → Disk Management.

2 See

The top shows volumes; the bottom shows each disk as a bar of partitions.

3 Do

Create, delete, format, shrink, extend and assign drive letters.

4 Careful

Deleting or formatting a volume erases its data.

Dynamic disk – volume types

1 Simple volume

Uses space on one disk, like a normal partition.

2 Spanned volume

Joins space from several disks into one large volume.

3 Striped volume

Splits data over disks for speed (RAID 0). No safety.

4 Mirrored volume

Keeps two copies on two disks (RAID 1). Safe if one disk fails.

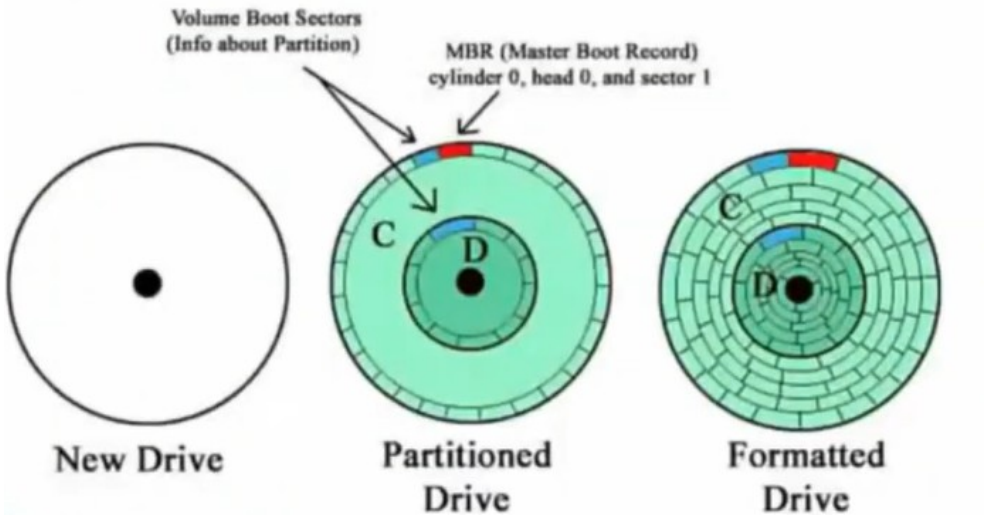
5 RAID-5 volume

Stripes data with parity; survives one disk failure (Server editions).

Hard disk Partitions

FROM THE LECTURE

- Primary
- Secondary
- Logical



Hard disk Partitions – Explained

1 Primary

Can hold an operating system and boot the computer. Drive C: is a primary partition.

2 Secondary (extended)

A container that is not bootable itself; it can hold logical drives.

3 Logical

Drives created inside the extended partition, such as D: and E:.

4 In the picture

New drive (empty) → Partitioned drive (MBR and volume boot sectors) → Formatted drive (ready for files).

Partitions – comparison

	Primary	Secondary / Extended	Logical
Role	Boots an OS	Container only	Drive inside extended
Max on MBR disk	Up to 4	1 (counts as one of 4)	Many
Bootable	Yes	No	Normally data only
Example	C:	Holds D:, E:	D:, E:

Why partition a disk?

1 Separate OS and data

Reinstalling Windows on C: keeps files on D: safe.

2 Organise

Keep work, movies and backups in separate drives.

3 Multiple systems

Each operating system can have its own partition.

4 Easier backup

You can back up or format one partition at a time.

Partitions Table

MBR

Master Boot Record. The older table, stored in the first sector of the disk.

GPT

Written as in the lecture. GPT = GUID Partition Table, the newer table used with UEFI.

Partitions Table – Explained

	MBR	GPT (GUID Partition Table)
Full name	Master Boot Record	GUID Partition Table
Disk size	Up to 2 TB	Beyond 2 TB (up to 9.4 ZB)
Partitions	4 primary (or 3 + extended)	Up to 128
Boot mode	Legacy BIOS	UEFI
Safety	One copy at the start	Backup copy at the end

MBR and GPT in detail

1 MBR layout

Sits in the first sector (cylinder 0, head 0, sector 1): 446 bytes boot code, 64 bytes for 4 partition entries and a 2-byte signature.

2 GPT layout

Has a protective MBR, a header, up to 128 partition entries and a backup copy at the end of the disk.

3 Which to use

Use GPT for disks over 2 TB and UEFI systems; MBR for old BIOS systems.

4 Check yours

Disk Management → right-click disk → Properties → Volumes → Partition style.

Format

1 What is format

Prepares a partition with a file system so Windows can store and find files.

2 Quick format

Rebuilds the file table fast; old data may still be recoverable.

3 Full format

Also checks the disk for bad sectors, so it takes longer.

4 Warning

Formatting removes the files on the partition. Back up first.

How to format a volume

1. Select

Open Disk Management and right-click the volume.

2. Format

Choose Format. Pick the file system (NTFS, FAT32 or exFAT).

3. Options

Set the volume label and allocation unit size; tick Quick Format.

4. Confirm

Click OK. All data on the volume will be erased.

File System

FAT

File Allocation Table. Simple and old; used on pen drives.

NTFS

The standard Windows file system with security and large files.

REFS

Written as in the lecture. ReFS = Resilient File System, for servers.

File System – Explained

	FAT	NTFS	ReFS
Meaning	File Allocation Table	New Technology File System	Resilient File System
File size	Small limits (4 GB in FAT32)	Very large files	Very large files
Security	None	Permissions and encryption	Data integrity and self-repair
Used on	Pen drives, old systems	Windows drives	Servers and big storage

File systems in detail

1 FAT

FAT12, FAT16 and FAT32 are very compatible. FAT32 has a 4 GB maximum file size; exFAT removes this limit.

2 NTFS

Supports permissions, encryption, compression, journaling and very large files. Default for Windows.

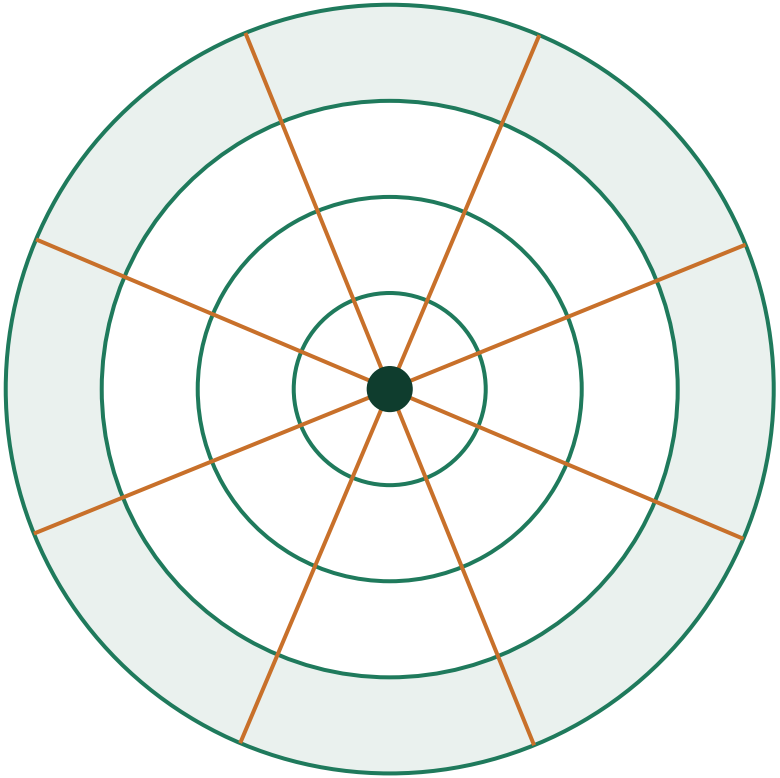
3 ReFS

Resilient File System: uses checksums to detect and repair corruption; made for servers and big storage.

4 Choosing

Pen drive for all devices: FAT32 or exFAT. Windows drive: NTFS. Server data: ReFS.

Tracks & sectors



Track
A circular ring on the platter. Data is written along tracks.

Sector
A track is divided into small slices called sectors (usually 512 bytes or 4 KB). A sector is the smallest unit the disk reads or writes.

MBR
Stored at cylinder 0, head 0, sector 1 – the first sector (from the lecture picture).

Tracks and sectors – more terms

1 Cylinder

The same track number on all platters, stacked like a cylinder.

2 Head / surface

Each platter side has its own read/write head.

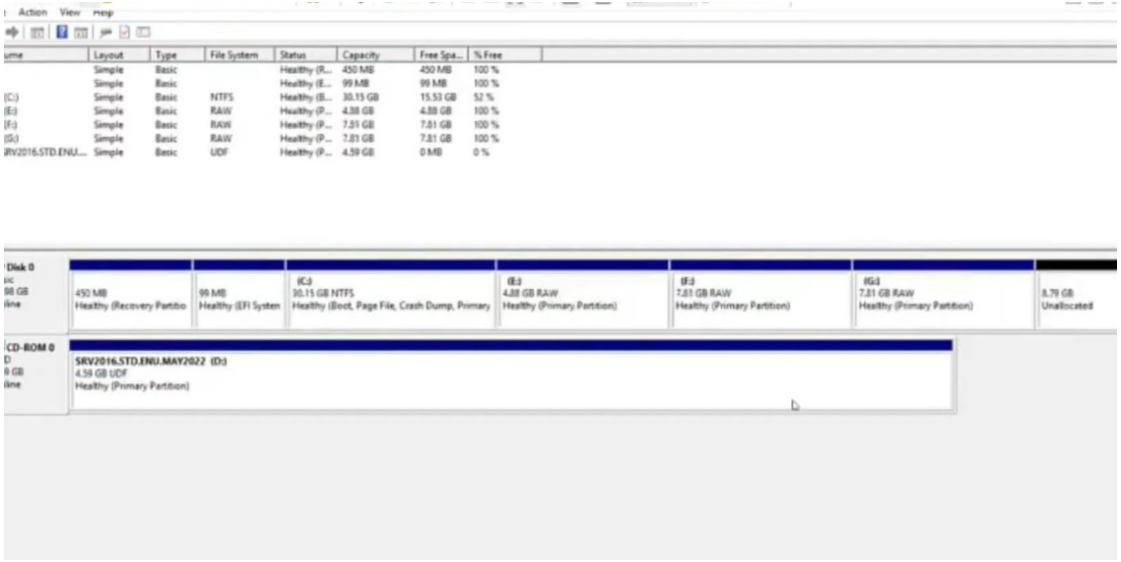
3 Cluster

A group of sectors; the smallest space a file system gives to a file.

4 Capacity

Cylinders × heads × sectors per track × bytes per sector.

Disk Management window



Disk 0 has a recovery partition (450 MB), EFI system (99 MB), C: 30.15 GB NTFS, three RAW partitions E:, F:, G: and 8.79 GB unallocated space. D: is the CD-ROM (UDF).

C: NTFS
Boot, page file and crash dump – the system drive.

RAW
No file system yet; format before use.

Unallocated
Free space not in any partition; can create or extend.

Recovery / EFI
Small system partitions; leave them alone.

Resize disk

Shrink

Right-click a volume in Disk Management → Shrink Volume. This frees space as unallocated.

Extend

Right-click → Extend Volume to add unallocated space. It needs unallocated space directly next to the volume. Back up data first.

Shrink and extend – steps

1 Shrink steps

Right-click the volume → Shrink Volume → enter the amount in MB → Shrink. New unallocated space appears.

2 Extend steps

Right-click the volume → Extend Volume → follow the wizard to add the unallocated space.

3 Limits

The unallocated space must be right next to the volume on the same disk.

4 Safety

Back up first. Unmovable files may limit how far you can shrink.

Thank you

Hard disk = permanent storage: platters + heads store data in tracks and sectors, organised by partitions and file systems.